

JAVA FULL STACK		
OBJECTIVE: The objective of this Java Full Stack Development course is to equip learners with a robust understanding and practical skills necessary to design, develop, and deploy dynamic web applications using Java technologies. By mastering HTML, CSS, JavaScript, Java, Spring Framework, and MySQL, students will be able to create end-to-end web solutions, effectively manage front-end and back-end development, and handle database interactions efficiently. This course aims to provide a strong foundation in web technologies, enhance problem-solving abilities, and promote best practices in software development.		
Duration:		
S.NO	CONTENT	HOURS (L/P)
Frontend Development (HTML, CSS, and JavaScript)		
	HTML	
1	Introduction to HTML What is HTML? HTML Text Editor	
2	HTML Basics HTML Tags HTML Attributes HTML Elements	
3	Text Formatting HTML Formatting HTML Heading HTML Paragraph	
4	Links and Images HTML Anchor (Links) HTML Image	
5	Tables and Lists HTML Table <u>HTML Lists</u> Ordered List Unordered List Description List	
6	Forms in HTML HTML Form Form Input Types Form Attributes	
7	Advanced HTML HTML with CSS HTML Classes and IDs HTML Iframes HTML JavaScript Integration HTML Comments	
8	HTML Layout and Responsive Design HTML Layout Basics HTML Responsive Techniques	
9	Miscellaneous HTML Color Names HTML Button onClick HTML Favicon HTML Login Form	

	CSS	
10	Introduction to CSS What is CSS? CSS Syntax CSS Selector Types	
11	Applying CSS Inline CSS Internal CSS External CSS CSS Comments	
12	CSS Properties Background Properties Border Properties B14Display Properties Font Properties Color and Text Properties Box Model (Margin, Padding, Width, Height) Positioning (Absolute, Relative, Fixed)	
13	Advanced CSS CSS Flexbox and Grid (justify-content, align-items) CSS Pseudo-classes CSS Transitions and Animations CSS Media Queries for Responsive Design	
14	Miscellaneous CSS Icons CSS Box-shadow and Text-shadow CSS List Styling CSS Navigation Bar Styling	
	JavaScript	
15	Introduction to JavaScript JavaScript Basics JavaScript Example	
16	Variables and Data Types JavaScript Variables JavaScript Data Types JavaScript Operators	
17	Control Flow Conditional Statements (if, switch) Loops (for, while)	
18	Functions and Objects JavaScript Functions JavaScript Objects JavaScript Arrays, Strings, Dates, Math, Numbers, Booleans	

19	DOM Manipulation Document Object Model (DOM) Accessing DOM Elements Manipulating DOM Elements InnerHTML vs InnerText	
20	Events and Validation JavaScript Events Overview Event Handling (addEventListener) Form Validation using JavaScript Email Validation Example	
21	Browser Object Model (BOM) Window Object History Object Navigator Object Screen Object	
22	Miscellaneous External JavaScript JavaScript Comments JavaScript Best Practices	
<u>Backend Development (Java Spring boot and MySQL)</u>		
	Java Programming	
23	Java Basics	
i	Introduction to Java What is Java? History of Java Features of Java C++ vs Java Hello Java Program Program Internal How to set path? JDK, JRE, and JVM JVM: Java Virtual Machine	
ii	Java Basics Java Variables Java Data Types Unicode System Operators Keywords	

iii	Control Statements Java Control Statements Java If-else Java Switch Java For Loop Java While Loop Java Do While Loop Java Break Java Continue Java Comments Java Programs	
iv	Java Object Class Java OOPs Concepts Naming Convention Object and Class Method Constructor Static keyword this keyword	
v	Java Inheritance Inheritance (IS-A) Aggregation (HAS-A)	
vi	Java Polymorphism Method Overloading Method Overriding Covariant Return Type Super keyword Instance Initializer block Final keyword Runtime Polymorphism Dynamic Binding Instance of operator	
vii	Java Encapsulation Package Access Modifiers Encapsulation	
viii	Java Arrays Java Array	
ix	Java String What is String Immutable String String Comparison String Concatenation Substring Methods of String class StringBuffer class StringBuilder class String vs StringBuffer StringBuffer vs Builder	

x	Java String Methods String charAt() String compareTo() String concat() String contains() String endsWith() String equals() String equalsIgnoreCase() String format() String getBytes() String getChars() String indexOf() String intern() String isEmpty() String join() String lastIndexOf() String length() String replace() String replaceAll() String split() String startsWith() String substring() String toCharArray() String toLowerCase() String toUpperCase() String trim() String valueOf()	
xi	Exception Handling Java Exceptions Java Try-catch block Java Multiple Catch Block Java Nested try Java Finally Block Java Throw Keyword Java Exception Propagation Java Throws Keyword Java Throw vs Throws Final vs Finally vs Finalize	
xii	Java JDBC JDBC Introduction JDBC Driver Connectivity with MySQL Statement ResultSet PreparedStatement	
Python MySQL		

24	Environment Setup Setting up MySQL server Installing MySQL connector	
25	Database Connection Connecting to MySQL database using Python	
26	Creating New Database Creating and selecting a database	
27	Creating Tables Defining and creating tables	
28	Insert Operation Inserting data into tables	
29	Read Operation Querying and reading data from tables	
30	Update Operation Updating existing records	
Java Spring boot		
31	Introduction to Spring Boot What is Spring Boot? Setting up the Environment Installing Spring Boot Your First Spring Boot App Understanding the Project Structure	
32	Spring Boot Basics a. Routing i. Creating routes for your application. b. URL Mapping i. Mapping URLs to controller methods. c. HTTP Methods i. GET, POST, PUT, DELETE. d. Template Rendering with Thymeleaf i. Rendering HTML templates. e. Static Files i. Serving CSS, JavaScript, and images.	
33	Templates and Forms a. Template Inheritance i. Using base templates. b. Thymeleaf Logic i. Using conditions and loops in templates. c. Web Forms i. Creating and handling forms. d. Spring Boot Form Binding i. Simplifying form handling. e. Form Validation i. Validating form inputs.	

34	Working with MySQL Connecting to MySQL Setting up Flask with MySQL. CRUD Operations Creating, reading, updating, and deleting records.	
35	User Authentication User Registration Creating new user accounts. Login and Logout Managing user sessions. Password Hashing Securing passwords. Session Management Maintaining user sessions.	
36	Working with Databases a. Connecting to MySQL i. Setting up Spring Boot with MySQL. b. CRUD Operations i. Creating, reading, updating, and deleting records.	
37	User Authentication a. User Registration i. Creating new user accounts. b. Login and Logout i. Managing user sessions. c. Password Hashing i. Securing passwords. d. Session Management i. Maintaining user session.	